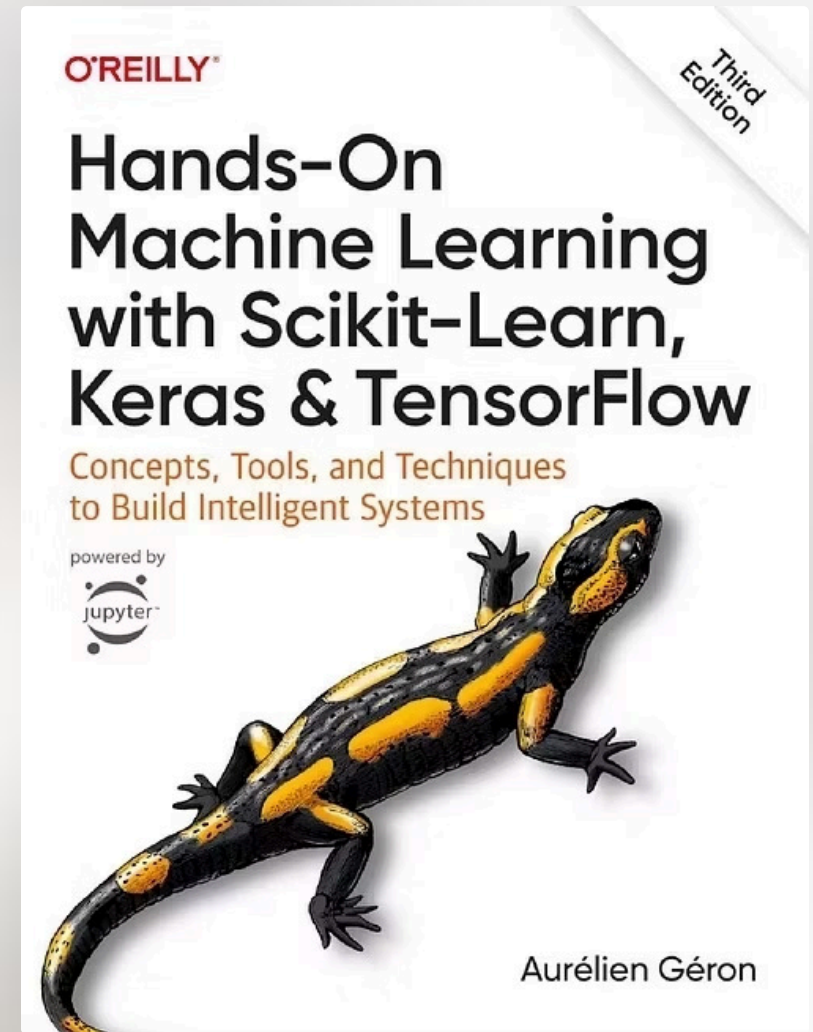


Machine Learning

Lessons based on the Textbook "Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow" by Aurélien Géron, 2023.

RP by Rodrigo Martins Pagliares

Last update: 07/08/2025



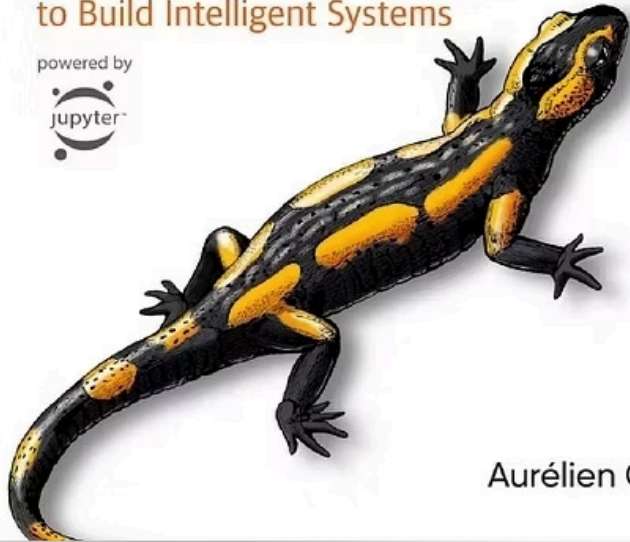
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Third
Edition

Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow

Concepts, Tools, and Techniques
to Build Intelligent Systems

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Aurélien Géron

Machine Learning

Chapter 1 – The Machine Learning Landscape

Agenda for Chapter 1

1. What Is Machine Learning?
2. Why Use Machine Learning?
3. Examples of Applications
- 4. Types of Machine Learning Systems**
5. Main Challenges of Machine Learning
6. Testing and Validating



A futuristic robot with a metallic, gold-colored body and glowing blue lights on its head, chest, and joints stands on a dark, rocky surface. In the background, a large planet with prominent rings, resembling Saturn, is visible against a dark, starry sky.

Agenda for Chapter 1

1. What Is Machine Learning?
2. Why Use Machine Learning?
3. Examples of Applications
4. **Types of Machine Learning Systems**
 - Classification criteria #1: Supervision during training
 - Supervised, Unsupervised, Semi-Supervised, Self-Supervised, and Reinforcement Learning
 - Classification criteria #2: Approach to Learning Method
 - Batch and Online (Incremental) Learning
 - **Classification criteria #3: Learning Method**
 - **Instance-Based** and Model-Based Learning
5. Main Challenges of Machine Learning
 - Insufficient Quantity of Training Data
 - Non-representative Training Data
 - Poor-Quality Data
 - Irrelevant Features
 - Overfitting the Training Data
 - Underfitting the Training Data
 - Stepping Back
6. Testing and Validating
 - Hyperparameter Tuning and Model Selection
 - Data Mismatch

Classification criteria #3: Learning Method

Instance-Based and **Model-Based** Learning

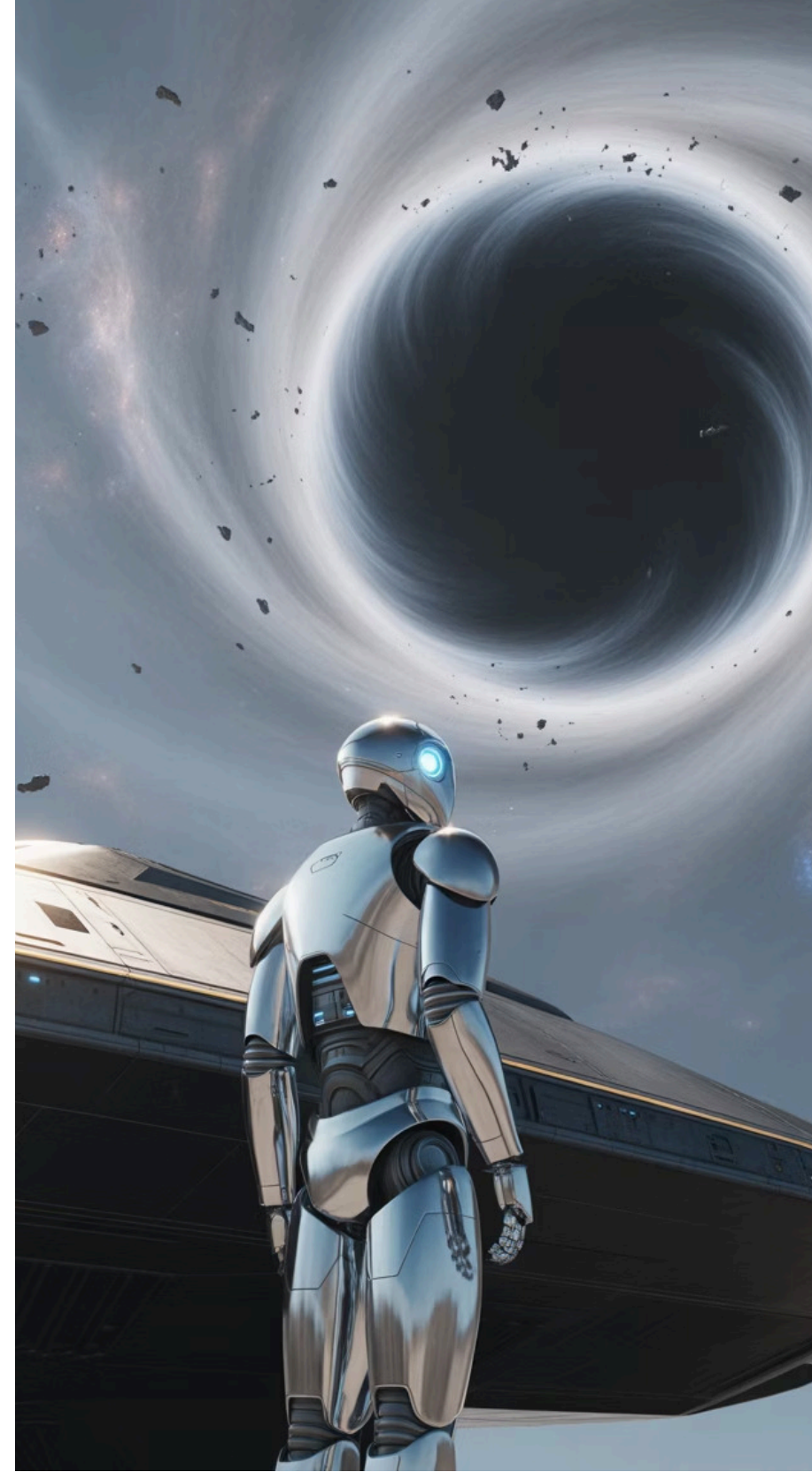
Instance-Based Learning: Overview

Definition:

- A type of ML where the **system generalizes** by **comparing new instances to previously seen examples**.
- Uses a **similarity measure** to make predictions based on the **most similar instances** in the training data.

How It Works:

- **Memorizes specific instances** from the training data.
- When a **new instance** arrives, it is **classified** based on its **similarity** to the **known instances**.
- **Example:** Spam filter flags emails similar to previously flagged spam.



Example: Spam Filter

1

Trivial Spam Filter (Learning by Heart)

- Flags emails that are **IDENTICAL** to **known spam emails**.
- **Limitations:** Doesn't work well for slightly altered or new types of spam.

2

Improved Spam Filter (Using Similarity)

- Flags emails that are **SIMILAR** to known spam, even if they are not identical.
- **SIMILARITY MEASURE:** Count the number of **common words** between two emails.
- **Outcome:** Emails with many common words to known spam are flagged.

Similarity Measure in Instance-Based Learning

0.78



Known Spam Email

Contains specific **words** and **patterns** that have been **identified** as **spam**



Basic Similarity Measure

Count the **common words** between two **emails**.



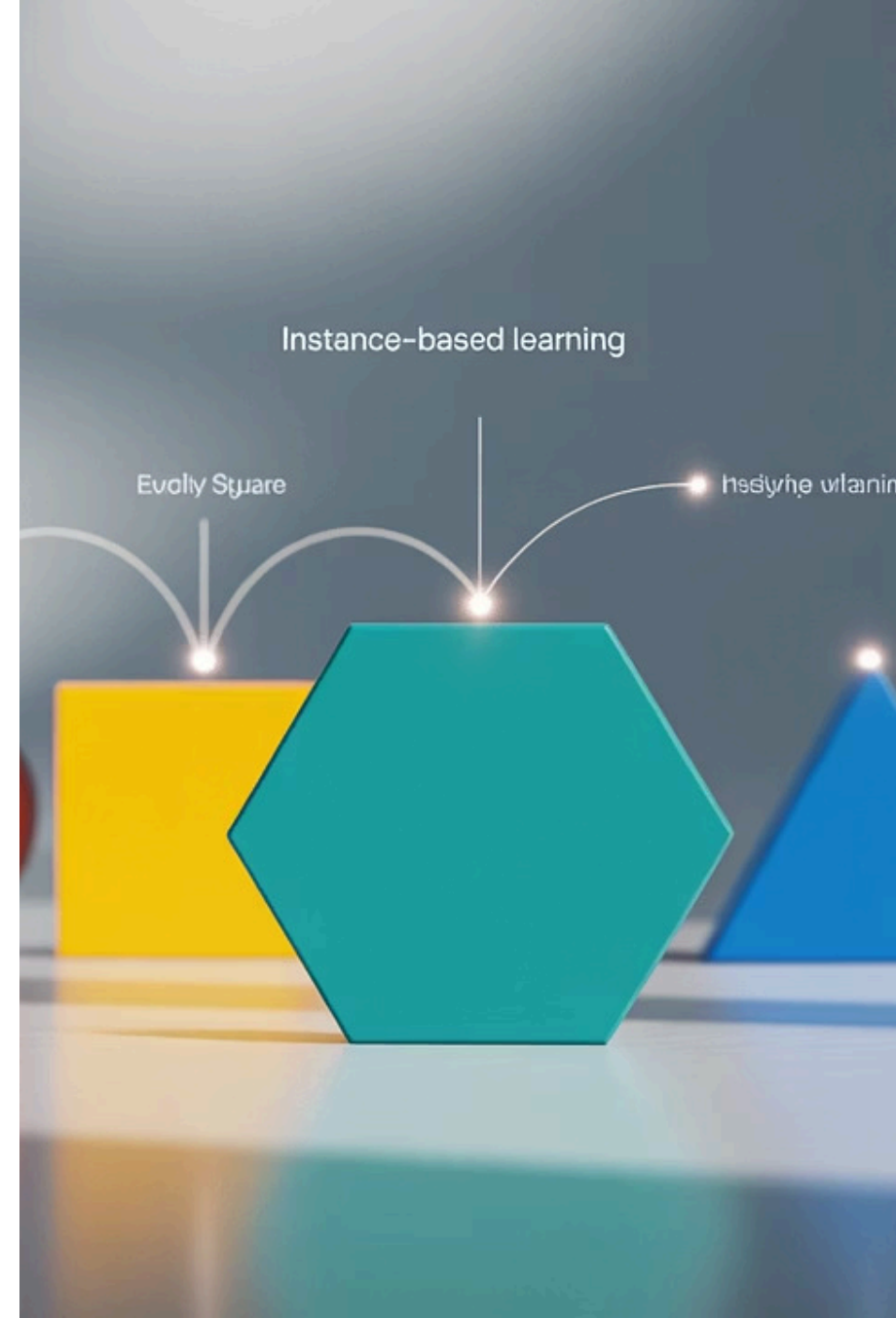
Classification

If a new email shares many words with a **known spam email**, it is **flagged** as **spam**.

Instance-Based Learning: Generalization

How the System GENERALIZES:

- **Learned examples** are used to classify **new instances** based on their similarity.
- **Example:** A new shape is classified based on the majority class of its most similar instances.



Example: Instance-Based Learning

In this example, the **new instance** would be classified as a **triangle** because the **majority of the most similar instances belong to that class**.

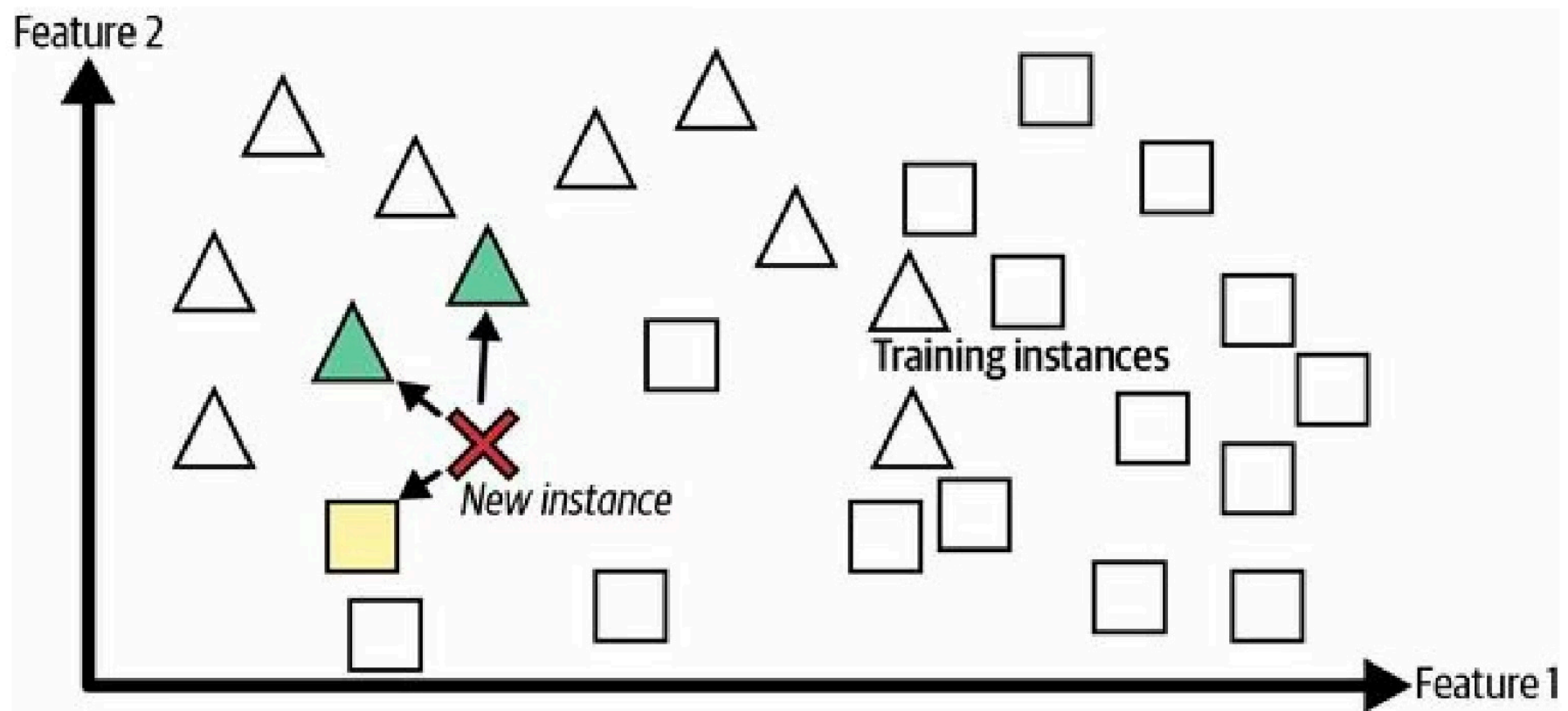


Figure 1-16. Instance-based learning

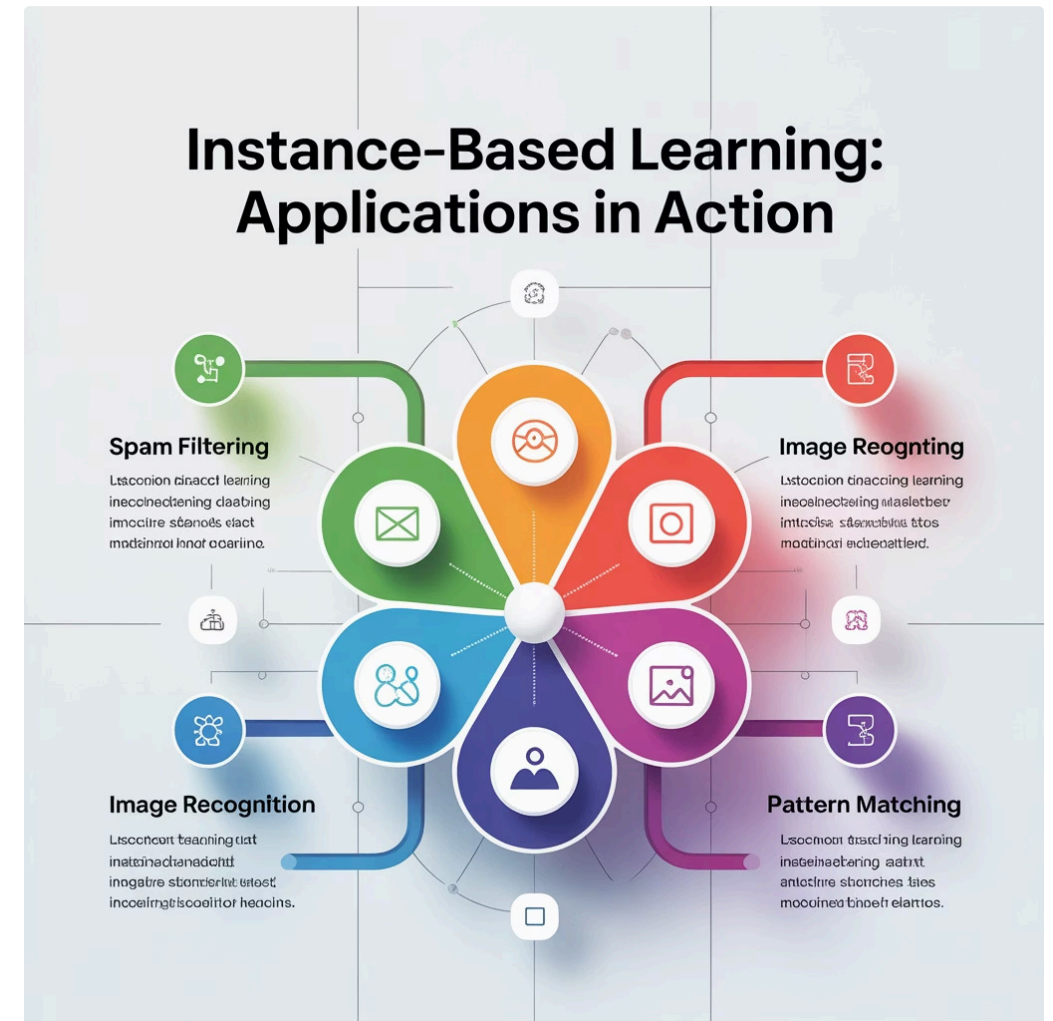
Key Takeaways

Instance-Based Learning

The system memorizes examples and generalizes to new cases using similarity measures.

Applications

Useful in spam filtering, image recognition, and other tasks requiring pattern recognition from existing data.



Next: Model-Based Learning and a Typical ML Workflow

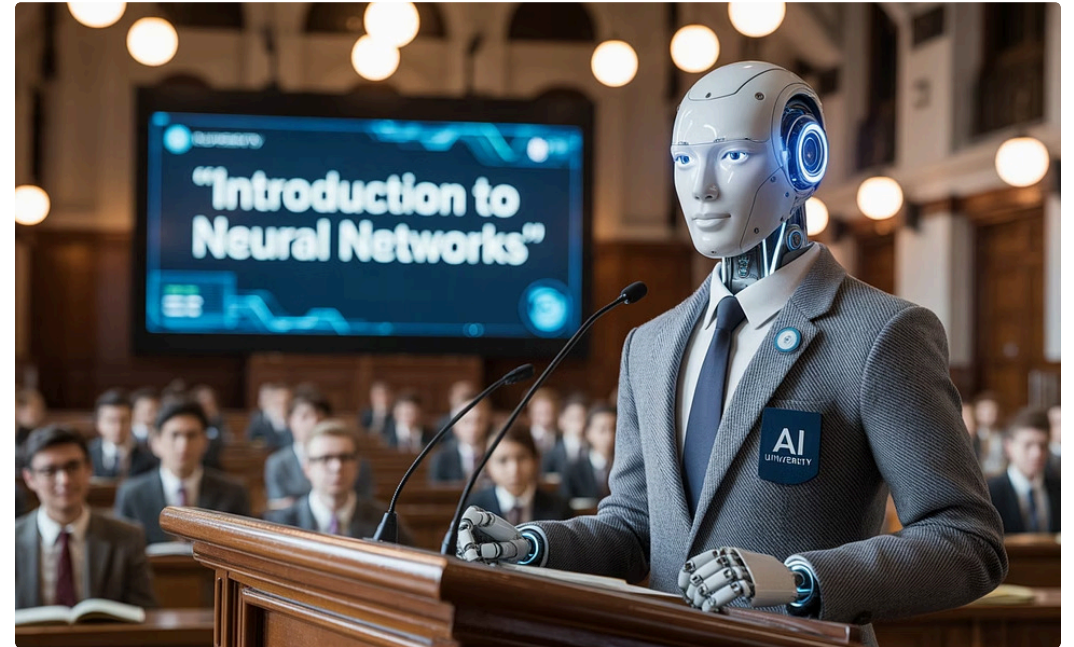


Q&A



Open Floor

Open floor for discussion or clarifications



Discussion

Share your thoughts and questions about AI concepts